

Chain-of-Thought Distillation for Drug-Drug Interaction Prediction: Transferring Pharmacological Reasoning from Large to Small Language Models

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1 Abstract

Large language models (70B+ parameters) can generate detailed chain-of-thought (CoT) reasoning about DDIs but remain impractical for real-time clinical deployment. This study focuses on a multi-stage knowledge distillation framework that transfers pharmacological reasoning from a large teacher model (Llama-3.3-70B-Instruct) to a student model (Qwen2.5-7B-Instruct) using the Drug-Bank DDI dataset. This represents the first application of CoT distillation to DDI prediction, a setting that differs substantially from prior CoT distillation work. Teacher-generated reasoning traces are quality-filtered by an ensemble of three independent judge models, yielding 144,217 high-quality training examples. The student is fine-tuned via LoRA using a sequential two-phase curriculum: first on classification labels to establish a strong baseline (Macro F1 = 0.9196), then on CoT traces to introduce pharmacological reasoning. Our experiments reveal a fundamental tension between reasoning and classification: naive CoT distillation degrades Macro F1 by 39.8 points due to token-level gradient imbalance, while sequential training recovers most accuracy (Macro F1 = 0.8475) but leaves a 7.2-point gap. We identify three root causes: compact CoT generation, classification weighted loss, and extreme class imbalance; and propose targeted interventions to address each.

2 Introduction

Drug-drug interactions (DDIs) contribute to 30% of adverse drug reactions, making accurate prediction critical for patient safety (De Vito et al., 2026). Traditional machine learning approaches struggle to capture the complex relationships between molecular structures and pharmacological outcomes (De Vito et al., 2026). Recent work has shown that language models can achieve strong performance in DDI classification by processing

drug names, their molecular structures, and textual descriptions (De Vito et al., 2026). However, these models are only able to state that the interaction will occur, they cannot explain why. Understanding the mechanistic basis of DDIs is essential for clinical decision-making, as two drugs that work through distinct pathways may compound to produce harmful effects, that may not have otherwise occurred.

Large language models (70B+ parameters) can generate detailed pharmacological reasoning, producing step-by-step explanations referencing specific enzyme isoforms, receptor binding sites, and metabolic pathways (Wei et al., 2022). However, deploying such models in clinical settings is impractical: a 70B model requires approximately 140GB of GPU memory and exhibits inference latency measured in seconds per prediction. This creates a fundamental dilemma: small models (7B parameters) are deployment-efficient but lack reasoning capability, whereas large models can reason but cannot be deployed at scale.

Recent advances in knowledge distillation (Hinton et al., 2015) and chain-of-thought prompting (Wei et al., 2022) suggest a potential solution. Magister et al. (2023) demonstrated that reasoning capabilities can be transferred from large models (100B+ parameters) to smaller models (220M–11B) through supervised fine-tuning on teacher-generated reasoning traces, achieving substantial performance gains on mathematical reasoning tasks. However, their work focused on arithmetic and commonsense reasoning with relatively balanced class distributions, never exploring biomedical applications or molecular prediction.

We investigate whether pharmacological reasoning can be transferred from a 70B teacher model to a 7B student model through CoT distillation for drug-drug interaction prediction. We find that while reasoning transfer is possible, naive distillation degrades classification performance, and we

identify the underlying causes and propose targeted solutions.

3 Related Work

Chain-of-Thought Prompting Wei et al. (2022) demonstrated that step-by-step reasoning examples in few-shot prompts dramatically improve LLM performance, boosting PaLM 540B accuracy on GSM8K from 17.9% to 56.9% (Min et al., 2022). Wang et al. (2022) and colleagues further improved CoT through self-consistency decoding across multiple reasoning paths. However, CoT reasoning is an emergent ability restricted to models exceeding 100 billion parameters, smaller models produce fluent but illogical reasoning that degrades performance. None of these works explored biomedical or molecular prediction tasks.

Knowledge Distillation Hinton et al. (2015) pioneered knowledge distillation, showing that a small student model can match a large teacher by training on soft probability distributions rather than hard labels, with soft targets encoding rich inter-class similarity information. Gou et al. (2021) survey distillation techniques across knowledge types including logits, features, and attention maps. However, traditional distillation transfers classification probabilities rather than reasoning processes, and was never applied to biomedical domains.

Chain-of-Thought Distillation Magister et al. (2023) demonstrated that CoT reasoning can be distilled from large models (100B+) to smaller models (220M–11B) via supervised fine-tuning on teacher-generated traces, improving GSM8K accuracy from 8.11% to 21.99% for T5-XXL (11B). Key findings include that answer-conditioned prompting improved trace quality by 19 percentage points, and filtering incorrect teacher outputs (retaining ~90% of traces) was essential for student performance. Ho et al. (2023) explored diverse sampling strategies, while Li et al. (2023) focused on symbolic reasoning with structured output formats. Our work follows the methodological framework of Magister et al. (2023), adopting their two-phase pipeline, but extends it to biomedical molecular prediction with extreme class imbalance — a setting never explored in prior CoT distillation work.

LLMs for Drug-Drug Interaction Prediction

De Vito et al. (2026) demonstrated that fine-tuned small LLMs achieve 91.9% accuracy on DDI classification by processing SMILES strings and drug

descriptions as text, with fine-tuning on 1,000 examples outperforming zero-shot and few-shot approaches (Boyce et al., 2012). Zhao et al. (2025) and colleagues developed a collaborative multi-LLM system generating evidence-based drug analysis, but requires deploying multiple large models simultaneously, limiting clinical feasibility. Both approaches achieve high classification accuracy but operate as black boxes, providing no mechanistic explanation of enzyme interactions or metabolic pathways.

4 Methodology

4.1 Dataset and Task Definition

We use the Drug–Drug Interaction (DDI) dataset from the Therapeutics Data Commons (Wishart et al., 2018), derived from the database DrugBank. DrugBank contains clinically documented drug interactions compiled from scientific literature, regulatory sources, and expert review, making it well suited for evaluating whether a model can learn and transfer mechanistic reasoning about drug interactions.

The dataset contains approximately 191,808 drug pairs annotated across 86 interaction types. Each instance includes two DrugBank IDs, the corresponding SMILES chemical representations for both drugs, and an integer label representing the interaction type. The ground truth therefore consists of multi-class interaction labels describing the relationship between two drugs.

This work formulates the task as a multi-class NLP classification problem. The input consists of textual prompts containing the drug names and their SMILES sequences, while the output is the predicted interaction type, preceded by generated reasoning. Since SMILES encode molecular structure as character sequences, the model processes sequential language representations of chemical structures.

A key limitation is severe class imbalance: the largest class contains 44,184 examples while the smallest contains only 5, producing an 8,837:1 ratio, with 24 of 86 classes having fewer than 50 training examples. We preserve this natural distribution as it reflects true clinical prevalence, though it introduces noise into Macro F1 evaluation.

4.2 Data Preprocessing

Raw dataset was loaded from a tab-separated values (TSV) file and column structure was standardized

with original columns renamed to consistent identifiers to simplify downstream processing. Data types were specified explicitly, with drug identifiers stored as strings and interaction labels as integers.

Next, drug identifiers were resolved into drug names for ease of readability. DrugBank IDs were mapped to their corresponding drug names. For identifiers that failed to map, the original DrugBank ID was retained. Moreover, two additional columns were added to the dataset to store the resolved names. A label mapping was also constructed to link each interaction class to its corresponding text template. The placeholders in these templates were then replaced with the resolved drugs names to produce the final label_text column.

The dataset was partitioned using stratified sampling to preserve class distribution across an 80/20 train-test split. The processed dataset was saved in JSON Lines format, with separate files for training and test sets and a label mapping file preserving interaction identifier-to-description relationships. Each record contains resolved drug names, DrugBank identifiers, SMILES molecular structures, interaction labels YY Y, and label-derived interaction descriptions, forming the structured input for prompt construction and model training.

4.3 Model Selection

We select **Llama-3.3-70B-Instruct** (Touvron et al., 2023) as the teacher model. Since its 4,096 token context length operates in a 9-shot prompting configuration with curated examples spanning diverse interaction categories. Generation uses temperature $\tau = 0.6$, top- $p = 0.95$, and max 1,024 output tokens. Inference is performed with vLLM (Kwon et al., 2023) using 4-way tensor parallelism in FP16.

Three models filter teacher CoTs for quality: (1) **Qwen2.5-72B-Instruct** (Qwen Team, 2025), a strong general-purpose model with broad scientific knowledge; (2) **DeepSeek-R1-Distill-Qwen-32B** (DeepSeek-AI, 2025), a reasoning-specialized model suited for evaluating logical chain coherence; and (3) **Mistral-Large-Instruct-123B**, providing an independent third perspective. Each judge evaluates every trace on five dimensions. A trace is retained only if at least two of three judges assign an overall score of ≥ 3 .

For the student model, we select **Qwen2.5-7B-Instruct** (Qwen Team, 2025), balancing capability with deployability. At 7B parameters, this model can run on a single GPU in clinical settings. We

fine-tune using LoRA (Hu et al., 2022) with rank $r = 64$, scaling factor $\alpha = 128$, and dropout 0.05. This yields 161.5M trainable parameters (2.08% of 7.78B total), keeping training efficient while providing enough capacity for both classification and reasoning.

4.4 System Pipeline

Stage 1: Following pre-processing, the dataset is stratified-split 80/20 into 153,446 training and 38,362 test pairs across 86 interaction types.

Stage 2: Llama-3.3-70B generates step-by-step reasoning for all training pairs. Each prompt includes a pharmacologist system instruction, the drug pair information, and nine curated few-shot examples. The teacher outputs reasoning traces averaging 306 words, each ending with a structured classification: “Classification: $Y=k$ — *interaction description*”.

Stage 3: All 153,446 teacher traces are evaluated by the three-judge ensemble. Traces passing majority consensus form the final CoT training set of 144,217 examples.

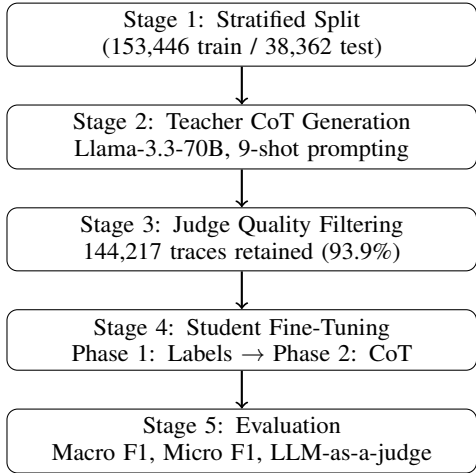


Figure 1: Pipeline overview: teacher CoT generation, three-judge quality filtering, and student fine-tuning.

Stage 4: Training proceeds in two sequential phases. Phase 1 trains on classification labels only for 3 epochs at learning rate 2×10^{-4} with cosine scheduling, a strong classification baseline. Phase 2 continues from the Phase 1 checkpoint on CoT data for 2 epochs at a $4\times$ reduced learning rate (5×10^{-5}), teaching the model to produce reasoning before classification. Both phases use 4-GPU DDP with effective batch size 64, BF16 mixed precision, gradient checkpointing, SFTTrainer, and max sequence length 2,560 tokens as well as

Temperature-scaled class resampling ($T = 0.5$, i.e., square-root sampling).

4.5 Compute and Feasibility

All experiments run on Runpod cloud infrastructure with 4×NVIDIA H100 80GB HBM3 GPUs (340 GB total VRAM), chosen to accommodate the memory requirements of the 70B teacher and three judge models during inference. Table 1 details the runtime per pipeline stage; the complete pipeline requires approximately 50–60 GPU-hours. All conditions are evaluated on the held-out 38,362-

Stage	Hardware	Time
Teacher CoT generation	4× H100	~18h
Judge ensemble evaluation	4× H100	~24h
Student training (per condition)	4× H100	~2h
Total		~50–60h

Table 1: Runtime breakdown per pipeline stage.

pair test set (20% stratified split). Macro-F1 serves as the primary metric given class imbalance, with Micro-F1 reported for comparability. Condition B serves as the classification upper bound baseline.

Teacher CoT generation requires ~18 hours using 4-way tensor parallelism, judge evaluation requires ~24 hours across three models, and student training requires ~2 hours per condition using 4-GPU DDP. Model checkpoints and cache reside on a high-performance scratch filesystem to avoid I/O bottlenecks. All experiments use seed 42 and a single YAML configuration file to ensure full reproducibility.

5 Evaluation Framework

The pipeline produces two distinct outputs, a predicted interaction class and a free-text pharmacological reasoning trace which requires separate evaluation methods. Classification correctness is assessed through standard multi-class metrics (Layer 1), while reasoning quality is evaluated through a rubric-based LLM-as-a-judge protocol (Layer 2). Conflating the two would obscure whether CoT distillation transfers interpretable reasoning or merely preserves label-prediction accuracy. We report both Macro-F1 and Micro-F1, following Sokolova and Lapalme (2009). Micro-F1 is reported for comparability with De Vito et al. (2026) but it is not our primary criterion, as a model failing entirely on rare classes can still report a misleadingly high score. Given the class imbalance

and the clinical consequence of missing rare interaction types, Macro-F1 is the primary metric.

Five experimental conditions are evaluated across label-only and CoT-based training configurations, with pairwise Macro-F1 deltas used to isolate the effect of each intervention. For reasoning quality, BLEU and ROUGE are insensitive to pharmacological accuracy (Papineni et al., 2002); we therefore adopt the LLM-as-a-judge paradigm (Zheng et al., 2023), with an independent three-judge ensemble scoring the five dimensions, including mechanism accuracy and causal chain quality. A stratified sample will be scored for the teacher and C_{seq} on identical inputs, over-representing the CYP confusion classes ($Y=47$, $Y=73$, $Y=49$) to determine whether the classification gap reflects routing failure or reasoning degradation—high CAUSAL_CHAIN and MECHANISM_ACCURACY on misclassified traces would confirm the former. Inter-judge agreement is reported via Krippendorff’s α (Krippendorff, 2011) ($\alpha > 0.60$), and reasoning transfer as the per-dimension score delta between C_{seq} and the teacher, where a non-negative delta indicates successful transfer on dimension d .

6 Experiments and Progress

6.1 Experimental Setup

Condition	Description
B	Label-only SFT; accuracy upper bound
C_{naive}	Full-length CoT from scratch; reproduces naive distillation failure mode
C_{seq}	Sequential curriculum: Phase 1 labels, Phase 2 CoT at 4× reduced lr
C_{mix}	50/50 interleaved label and CoT in a single training run
D_{real}	Single-pass generation from C_{seq} checkpoint; evaluates classification accuracy

Table 2: Training conditions.

All conditions share identical hyperparameters with 3uses the 38,362-pair test set with near-greedy decoding (temperature 0.1, top-p 0.9).

6.2 Experiment 1: Baseline Analysis

Condition B achieves Macro F1 of **0.9196** and Micro F1 of 0.9344 on the 86-class DDI task, with 100% valid label extraction and 85 of 86 classes predicted. Its outputs average only 15.8 words (just

the classification statement). This strong result confirms that Qwen2.5-7B with LoRA is highly capable at DDI classification when the training objective focuses exclusively on labels.

Condition C_naive drops to Macro F1 of **0.5542**—a catastrophic 39.8 point Macro F1 drop. Despite generating pharmacologically relevant reasoning (mean response length 214 words), the model’s ability to select the correct interaction type is severely degraded. The root cause is a fundamental **token-level gradient imbalance**. Teacher CoTs average 306 words of reasoning followed by only about 16 words of classification, creating a 94.9%/5.1% token split. Since the standard language modeling loss distributes gradient updates proportionally across all output tokens, 95% of the training signal pushes the model to generate better reasoning while only 5% teaches it to classify correctly.

6.3 Experiment 2: Improved Methods

Condition C_seq recovers classification accuracy to Macro F1 of **0.8475** (+29.3 over C_naive), but a 7.2-point gap to B remains. A pairwise disagreement analysis between B and C_seq (Table 4) shows that C_seq loses 3,704 examples that B classifies correctly while only recovering 831 that B gets wrong—a net loss of 2,873 examples. These losses concentrate heavily in three pharmacologically related classes: Y=47, Y=49 and Y=73. These classes share overlapping CYP450 enzyme mechanisms. For instance, CYP inhibition simultaneously decreases drug metabolism *and* increases serum concentration—and the verbose 300-word reasoning provides ample opportunity for the model to reason itself into a pharmacologically adjacent but incorrect class.

Condition C_mix yields only Macro F1 of **0.6715**, below C_seq. Interleaving label-only and CoT examples creates conflicting output format expectations, confirming that a sequential curriculum is preferable to simultaneous multi-objective training. Condition D_real initially produced catastrophic results (0.1832) due to an out-of-distribution prompt in Stage 2; after correction to single-pass generation from the C_seq checkpoint, it converges to C_seq levels.

6.4 Preliminary Results and Figures

Tables 3 and 4 together paint a clear picture: CoT distillation consistently degrades classification accuracy. Naive distillation loses nearly 40 Macro

Condition	Macro F1	Micro F1	CoT
B (Label-Only)	0.9196	0.9344	✗
C_naive	0.5542	0.7018	✓
C_mix	0.6715	0.7849	✓
C_seq	0.8475	0.8596	✓

Table 3: Classification on 38,362 DDI test pairs (86 classes). (C_seq) recovers most of the (C_naive) losses but trails B by 7.2 Macro F1 points

Pairwise disagreement (B vs. C_seq)	
B correct, C_seq wrong	3,704
B wrong, C_seq correct	831
Both correct / Both wrong	32,138 / 1,679
Net loss	2,873
Top confusion pairs in C_seq	
Y=47→Y=73 (metab ↔ serum conc)	551
Y=49→Y=47 (adverse ↔ metab)	321
Y=47→Y=49 (metab ↔ adverse)	306
Y=73→Y=47 (serum conc ↔ metab)	289
Y=49→Y=73 (adverse ↔ serum conc)	204

Table 4: Error analysis of C_seq. (C_seq) errors cluster by CYP450 mechanism (2,873 net loss).

F1 points, and even sequential training leaves a 7.2-point gap driven by 2,873 net lost examples concentrated in pharmacologically similar classes.

Our analysis identifies three root causes for this trade-off. First, the **token-level loss imbalance**. Second, **excessive CoT length** leading to verbose elaboration that overwhelms the 7B student’s capacity. Third, **extreme class imbalance** where the 8,837:1 ratio means rare interaction types receive almost no training signal; temperature sampling at $T=0.5$ only reduces this to approximately 94:1, far from adequate.

Based on these findings, we propose three targeted interventions. First, **Compact CoT** truncates teacher reasoning to approximately 100 words, shifting the token ratio from 95/5 to 86/14 while retaining the core pharmacological content. Second, a **Classification-weighted loss** applies a 10× multiplier to the cross-entropy loss for tokens following the “Classification” marker, an effective gradient split of 39% reasoning / 61% classification. Third, **Aggressive class balancing** lowers T to 0.25 and enforces a minimum of 50 examples per class, reducing the effective imbalance from 94:1 to approximately 10:1. These fixes are implemented as conditions C_wt (weighted loss only) and C_compact (all three combined); training is underway.

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